

EXERCISE I

In the context of deep neural networks

I.a) explain the concept of Loss function.

Multiclass SVM loss:

Given an example (x_i, y_i)
 where x_i is the image and
 where y_i is the (integer) label,

and using the shorthand for the
 scores vector: $s = f(x_i, W)$

the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \begin{cases} 0 & \text{if } s_{y_i} \geq s_j + 1 \\ s_j - s_{y_i} + 1 & \text{otherwise} \end{cases}$$

$$= \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

I.b) Consider the multiclass SVM loss (right).

Compute the loss value for a dataset with four samples and a classification problem with four classes C0, C1, C2, C3. Sample S0 has label C0, and its label scores are (1.1, 2, 0.1, 0.5), S1 has label C3 and its class scores are (2, 1.5, 0.1, 3), S2 has label C1 and its class scores are (2, 3, 1, 1), S3 has label C2 and its class scores are (0.2, 3, 1, 2)

	Sample 0	Sample 1	Sample 2	Sample 3
Class 0	1.1	2	2	0.2
Class 1	2	1.5	3	3
Class 2	0.1	0.1	1	1
Class 3	0.5	3	1	2

- For sample S0 with label C0 and scores (1.1, 2, 0.1, 0.5):

- $L_0 = \max(0, 2 - 1.1 + 1) + \max(0, 0.1 - 1.1 + 1) + \max(0, 0.5 - 1.1 + 1)$

- $L_0 = \max(0, 1.9) + \max(0, 0) + \max(0, 0.4)$

- $L_0 = 1.9 + 0 + 0.4 = 2.3$

- For sample S1 with label C3 and scores (2, 1.5, 0.1, 3):

- $L_1 = \max(0, 2 - 3 + 1) + \max(0, 1.5 - 3 + 1) + \max(0, 0.1 - 3 + 1)$

- $L_1 = \max(0, 0) + \max(0, 0) + \max(0, 0)$

- $L_1 = 0 + 0 + 0 = 0$

- For sample S2 with label C1 and scores (2, 3, 1, 1):

- $L_2 = \max(0, 2 - 3 + 1) + \max(0, 1 - 3 + 1) + \max(0, 1 - 3 + 1)$

- $L_2 = \max(0, 0) + \max(0, 0) + \max(0, 0)$

- $L_2 = 0 + 0 + 0 = 0$

- For sample S3 with label C2 and scores (0.2, 3, 1, 2):
 - $L_3 = \max(0, 0.2 - 1 + 1) + \max(0, 3 - 1 + 1) + \max(0, 2 - 1 + 1)$
 - $L_3 = \max(0, 0.2) + \max(0, 3) + \max(0, 2)$
 - $L_3 = 0.2 + 3 + 2 = 5.2$

So, with ($\Delta = 1$), the updated loss values for the given samples using ASCII characters are:

- $L_0 = 2.3$
- $L_1 = 0$
- $L_2 = 0$
- $L_3 = 5.2$

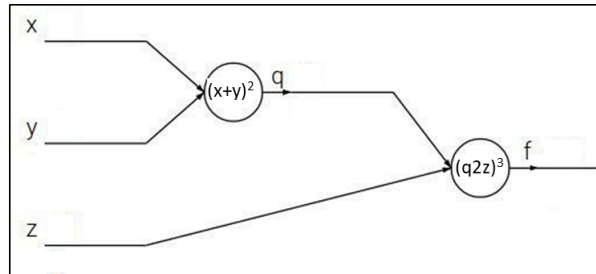
$L_{tot} = 7.5$

I.c) give an example of class scores for C0, ...C3 for which the SVM loss has value 0.

EXERCISE II

Consider the following data flow graph representing the function to the left:

$$f(x, y, z) = ((x + y)^2 2z)^3$$



II.a) Compute the partial derivatives on all edges of the nodes using backpropagation

II.b) compute the value of the gradients for $(x,y,z)=(1, 2, 3)$

II.c) compute the gradient in the same point as II.b when $q=x-2y$ and $f=z-2q$

EXERCISE III

A parallel cluster of RISC-V processors is executing a convolutional layer with 8 input channels and 19 output channels. Input spatial size is 61x61. Inputs, outputs and weights are quantized to 8 bits. Convolutional filter size is 5x5, padding with zeros is used, stride=2, no bias. RELU non-linearity is applied in place on the output. After RELU, results are average-pooled (2x2 pooling window, stride=2). Assume that on average a RISC-V processor can perform a MAC in 4 cycles and a RELU operation in 6 cycles and a division in 12 cycles). The three kernels (convolution, relu, average pooling) are parallelized over the output channels. No parallelization overhead

III.a) Compute the L1 minimal shared memory size that is needed to store the complete layer (input and output channels as well as weights) in L1

III.b) Compute the total number of cycles to process the layer on a 4-core cluster

III.c) Compute the speedup obtained when processing the layer with 4 cores versus only 1 core (sequential).